Machine Learning Course Project

Exercise Posture Classification System

Using Convolutional Neural Networks and MobileNetV2

Submitted by

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Submitted to

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Certificate

This is to certify that the project report titled Exercise Posture Classification System Using Convolutional Neural Networks and MobileNetV2 has been prepared and submitted by Chareesh Nara, matriculation number 5925155 for the course project of Machine Learning. This document is a report on the research work done for developing an image classification system using convolutional neural network on publicly available yoga posture image data set. It includes preprocessing, development, training, evaluation and interpretation of the results of the process. The research demonstrates the practical application of concepts including supervised learning, convolutional neural networks, transfer learning, and model evaluation. To the best of my knowledge, the report has been prepared in accordance with the requirements of the course and in the given academic report format. The results and findings described in this report have been produced by implementation in the Kaggle Notebook environment.

Name: Raja Hashim Ali

DECLARATION

I would like to say this project report titled "Exercise Posture Classification System Using Convolutional Neural Networks and MobileNetV2" is a completely new effort made by me as a part of Machine Learning course project. The data set used for this analysis is openly available on Kaggle and the experiment was conducted using Python, TensorFlow, Keras, and Kaggle Notebook. The results, figures, tables and observations contained in this report are the outcome of the project implementation and model validation by me. Also, I like to mention that I have taken due reference of all the resources used in this project. This report has not been submitted before for any other academic assignment. **Student Name:** Chareesh Nara

Matriculation Number: 5925155

Date: June

Acknowledgment

First, I would like to express my sincere gratitude to my instructor in Machine Learning for his assistance and helpful comments during this project. This project gave me the opportunity to learn about the process of working with deep learning techniques on a real world image classification problem. Secondly I would like to thank Kaggle for the Yoga Pose Image Classification Dataset and the Kaggle Notebook where this project was developed. The availability of open datasets and the notebook enabled the design and assessment of the posture recognition system with CNNs. Finally I would like to thank the University of Europe for Applied Sciences for the educational background which has enabled me to work on this machine learning project. The project has enhanced my knowledge of supervised learning, convolutional networks, transfer learning, evaluation, confusion matrices and technical research writing.

CONTENTS

List of Figures

List of Tables

Exercise Posture Classification System Using Convolutional Neural Networks and MobileNetV2

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Abstract

The recognition of exercise posture has become an important research area in computer vision and deep learning, because of its wide range of applications in intelligent fitness systems, rehabilitation, health care monitoring, and virtual training. Proper body posture while doing physical exercises is a requirement to prevent injuries and improve the efficiency of exercise. People who work out at home and do yoga alone may not always have access to the professional help that is needed to manually assess body postures. This is the reason why automatic posture recognition has gained a lot of interest inasmuch as it allows receiving real-time feedback without requiring constant human presence. Recent advances in Convolutional Neural Networks have significantly improved the performance of image classification tasks by automatically learning discriminative visual features from images. However, the classification of various yoga poses is challenging due to the similarity in body pose structure and very minor visual differences. This paper presents classification of exercise postures using CNN based on Yoga Pose Image Classification Dataset downloaded from kaggle. The dataset contains 5,993 RGB images, belonging to 107 posture classes. The images were resized to 224 x 224 pixels and preprocessed. We considered two deep learning models including custom Convolutional Neural Network and MobileNetV2 architectures. The implementation of both the models is done in Kaggle Notebook using TensorFlow and Keras. The performance of the models has been evaluated using parameters such as accuracy, loss curves, confusion matrix, precision, recall, F1 score and classification report. From the experimental results it can be seen that the problem of posture classification on 107 classes of similar looking images is a difficult problem owing to low number of images and high similarity between the classes. However, the designed system successfully demonstrates the process of image classification, including preprocessing, development of CNN models, training, evaluation, and visualisation of the results.

Index Terms

Convolutional Neural Network, image classification, yoga pose recognition, MobileNetV2, deep learning, posture classification

I. Introduction

Artificial intelligence has revolutionised many fields of research by enabling computers to detect patterns in complex data. Computer vision is one of the most prominent areas of Artificial Intelligence in which the machines are taught to analyse and understand visual information, among other things. Image classification is one of the major computer vision tasks where the machine classifies the input image into one of the predefined categories . Image classification is a common task in medicine, sports, autonomous vehicles, agriculture, industry, surveillance, fitness and many other areas. In the last few years, deep learning has been employed to improve image classification performance, and it doesn't need handcrafted features.

Posture identification of exercise is an important task in computer vision. Right posture helps you to train effectively and safely. If you're doing yoga or other exercises, the wrong posture can make the training less effective and can even cause injuries. Many people practise yoga and exercises by themselves without the guidance of a coach. In this case, automatic methods need to be developed to recognise exercise posture from images and provide some useful information.

Since features are learned from images, postures can be classified using images with Convolutional Neural Networks. CNN can recognise low level features such as edges, curves and texture. The higher layers of CNN learn more advanced features such as body shape, posture shape, and posture orientation. Thus, the use of CNN is much more effective as compared to the conventional machine learning techniques that rely much on the manually extracted features of the images. CNN models have been applied to medical imaging, object detection, face recognition, and human activity recognition.

Another technique that can be used in computer vision is called "transfer learning". Instead of training a full model from scratch, transfer learning utilises the existing framework which was designed to classify images. The commonly used frameworks for transfer learning are MobileNetV2, ResNet, VGG and EfficientNet. MobileNetV2 is especially useful due to its lightweight and fast computational nature. It classifies images and has less parameters than other bigger frameworks. Thus, MobileNetV2 was selected for use in this project.

The dataset used in this project has 5993 pictures of 107 different yoga pose classes. This is a difficult task since there are many classes and some yoga postures look the same. Indeed, some poses can be differentiated from each other by small modifications such as the position of the arms, the direction of the legs, the rotation of the body or the balance construction. If there are too many classes that look alike in the dataset, then the network has to be able to distinguish between these classes by looking at small visual differences. This becomes more difficult when there are few images per class.

The primary aim of this project is to develop a deep learning-based exercise posture classifier. TensorFlow and Keras aided us in designing both a custom CNN model and a model based on MobileNetV2. The dataset was downloaded from Kaggle, it was preprocessed (resized, normalised and augmented) and divided into the training and validation datasets. Both models were evaluated based on their accuracy, loss, confusion matrix, precision, recall and F1 score.

II. Background and Motivation

Exercise is now a major part of health, lifestyle changes and healing from injuries. Yoga is one such exercise that is done by people of all age groups due to its contribution towards flexibility, balance, breathing and psychological well-being. However, yoga is very much dependent on proper body positioning because incorrect posturing leads to reduced efficiency of the activity and possible pain or injury in the future.

Traditional postural correction requires the presence of an instructor. The system works well, but it is not always available to everyone. For instance, many people do yoga at home by means of videos from the internet or applications from mobile devices. So they will never come to know whether they are doing yoga in the right posture or not.

Machine learning provides a practical way to do this. The posture classification system can be developed by training the machine learning algorithm with labelled images with different postures. Once trained, the machine learning model will be able to classify the postures. This is a big step towards building a more complex posture correction system. To correct posture, you first have to recognise posture.

This is because the motive of the project is both practical and theoretical. The motivation behind this project in practice is to learn how deep learning can help fitness technology. Theoretical motivation for the project is the desire to understand how a CNN model can be built, tested and analysed in a real scenario. In addition to this, it is also worth mentioning that real-world datasets often have problems that are not encountered in textbook scenarios.

III. Related Work

One of the most explored topics in computer vision is image classification. The traditional image classification methods heavily relied on feature extraction techniques like edge detection, colour histograms, textures and shape features. Such approaches were domain specific and failed when images had different lighting or background. This made things easier with the concept of deep learning where model learns the features from the image automatically.

CNNs have been proved effective for various image recognition tasks. CNNs learn simple features such as lines and edges in the first layers. In the intermediary layers it recognises complex forms and parts of objects. Deep layers identify high level semantic features. This hierarchy of learning is what makes CNNs so good at visual pattern recognition. This hierarchy can also be useful for posture recognition as CNNs can recognise body shape and posture configurations.

Recognition of human activities is highly related to posture classification Human activity recognition typically refers to the recognition of movements like walking, running, jumping, and sitting. On the other hand, the classification of postures usually consists in recognising the human body posture in still images. In the recognition of yoga postures there are classes that can be visually very close. For example, two standing poses could have legs at the same angle, but the arms would be in different positions.

Because it can cut down on the need for huge amounts of task-specific datasets, this method has become popular. Since they have been trained on visual data in larger sets, the pretrained models will be able to perform high-quality feature extraction. MobileNetV2 is a lightweight and good-performing architecture that consists of depthwise separable convolutions and inverted residuals.

In this research work, we have compared custom CNN and MobileNetV2. The custom CNN has been taken as baseline model while MobileNetV2 has been taken as more complex model for comparison . A comparison between these two models has been done to check if a lightweight deep learning architecture performs better than a simple CNN.

IV. Gap Analysis

Much success has been achieved in the realm of CNN-based image classification but posture recognition in exercise poses brings its own set of challenges. In general, image classification datasets have visual classes that are easy to distinguish as those classes can refer to animals, vehicles, objects or scenes. But in yoga classification postures can be very similar in terms of physical structure. A small change in the position of the arms, legs or the spine may mean a different class.

The other disadvantage is that there are few images for each class. The dataset used in this study is 5,993 images in 107 posture classes. This means that the number of images per class is small. Deep learning algorithms tend to do a lot better if you have a lot of images in each class. If the number of classes is large and each class contains a small number of images the algorithm cannot learn enough class-specific features.

TABLE I
SUMMARY OF RELATED APPROACHES

Area	Method	Main Contribution
Traditional image classification	Handcrafted features	Manually extracted image features such as edges, texture and shape descriptors.
CNNs for classification	Convolutional Neural Networks	Automatic extraction of features from image pixels to be used for classification. Transfer learning
MobileNetV2, ResNet, VGG	Utilisation of existing deep architecture to enhance the model design.	
Human activity recognition	CNN and temporal models	Identification of physical activities and movement patterns of the body.
Classification of visually similar human body poses; Pose-based models; CNN; Yoga pose recognition		
Proposed work	Custom CNN and MobileNetV2	Multi-class classification and analysis of 107 classes of yoga poses.

Another problem is the variability of the postural images encountered in real life. The images can have different backgrounds, lightings, camera angles, attires and body types. The number of parameters is so large that it's very hard for the model to concentrate on the posture structure. A good posture recognition system should be invariant to these visual variables .

V. Problem Statement and Research Questions

Problem statement : The problem statement for this project is to classify exercise and yoga postures automatically from RGB images. Given an image as input, the system predicts the corresponding posture class . Since there is only one class label for each image, it can be considered as a supervised and multiclass classification problem. Classification problem is difficult due to the large number of classes (107) in the dataset.

This study is guided by the following research questions:

1) Can a customised Convolutional Neural Network classify images of yoga postures into appropriate posture categories? Difference between MobileNetV2 based model and custom CNN model?
   What classification problems can be identified from the confusion matrix and classification report? What is the effect of the large number of classes on the classification performance?
2) In the future, the following improvements may be considered to enhance the performance of the system:

VI. Purpose and Contributions

The goal of this project is to develop a deep learning exercise pose classifier. The goal of this project is to investigate the complete process of a real-world machine learning application. This includes data set selection, pre-processing, model design and training, and result interpretation.

The first is to build a data loading and preprocessing pipeline using TensorFlow and Keras. The second one is to design a custom CNN based model for the multi-class yoga pose recognition purpose. The third one is to build a comparison model with MobileNetV2. The fourth is the evaluation of both models through accuracy and loss curves, confusion matrices, precision, recall and F1 score. The fifth one is the analysis of practical problems in large scale classification of yoga poses.

VII. Description of the Dataset

The dataset used for this study is the Yoga Poses Image Classification Dataset from Kaggle. The dataset was selected as it has real image samples of yoga poses that can be used for image classification task. The data set is structured such that each folder contains one pose class. The folder structure allows the automatic labelling of classes by TensorFlow during the data reading. The dataset contains 5993 images in 107 pose classes.

A. Dataset Features

The data set contains a large number of categories of yoga poses. For instance, some of the classes are standing poses, sitting poses, balancing poses, forward bends, back bends, inverted poses. Therefore, the data set is appropriate for benchmarking the performance of the deep learning models on a practical problem of posture classification. But the problem is not without its difficulties. Some of the classes differ from each other, while others are very similar.

TABLE II
DATASET DESCRIPTION

Property	Value
Dataset name	Yoga Pose Image Classification Dataset
Source	Kaggle
Total images	5,993
Number of classes	107
Image type	RGB images
Input image size	224x224 pixels
Task type	Multi-class image classification
Training images	4,795
Validation images	1,198

And, there are natural differences in the visual images. For example, different background settings, clothing styles, body types, camera positions and lighting conditions. This is important as in real life a posture recognition system will not only work in one kind of environment. But all these factors contribute to the noise in the learning phase. The model should ignore all information except posture related information.

Fig. 1. Sample images from the Yoga Pose Image Classification Dataset.

B. Dataset Split and Distribution

The dataset was split into training and validation datasets with a partition ratio of 80:20. The model was trained using the training dataset by adjusting the parameters. The model was validated on the validation dataset to evaluate its performance on unseen data. The validation dataset is needed because the model accuracy on training data alone doesn't show how well the model will perform on new data.

TABLE III
TRAINING AND VALIDATION SPLIT

Subset	Images	Percentage
Training set	4,795	80 percent
Validation set	1,198	20 percent
Total	5,993	100 percent

For multi-class classifications, dataset's distribution is crucial. If there are lots of images for one set and very few images for another class set, then the model learns better representation of the class with more images. The distribution of dataset was done by counting class folders.

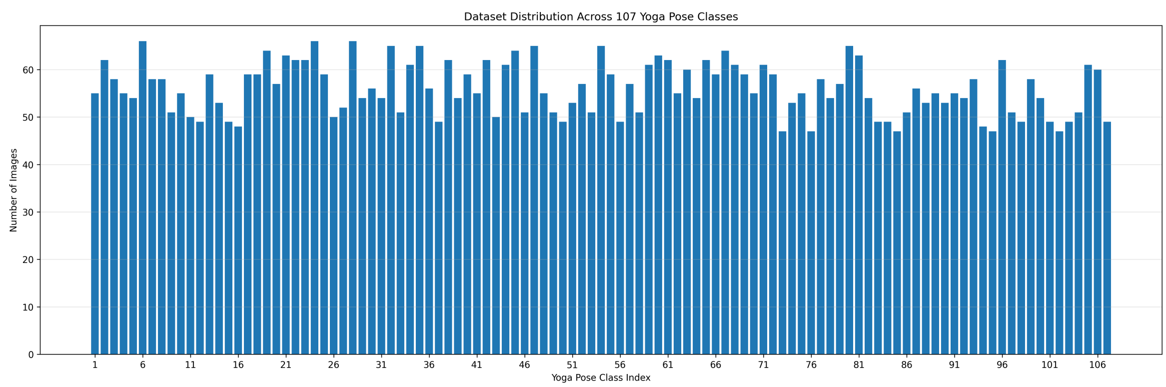

Fig. 2. Distribution of images among the 107 yoga posture classes.

VIII. METHODOLOGY

A. Data Loading

TensorFlow loaded the data from the folder directory. The `image_dataset_from_directory` function loads the images, assigns the label based on the folder name, resizes the image, and batches. This is useful as the dataset is already structured in the form of class folders. We ran the function with a fixed seed for randomness.

1) Image resizing and normalisation: The images were downsized to the size of 224x224 pixels. It is necessary to resize them because the input images need to be of the same size before being fed into the Convolutional Neural Network model. Images of different sizes cannot be directly fed into the CNN architecture.

The pixel values ranging from 0 to 255 were normalised to the range between 0 and 1. This was achieved through the rescaling layer in CNN architecture. Normalisation makes neural networks easier to train since all the input values are scaled down to a lower numerical value.

B. Data Augmentation

Data Augmentation technique was adopted to improve the performance of the model. The dataset has many classes with few images. Without augmentation, the model would tend to overfit the training data. Data augmentation is the process of adding random transformations to the training images during model training.

C. Overall System Workflow

The exercise posture classification framework is proposed as an end-to-end deep learning-based approach. The first step is gathering the Yoga Pose Image Classification Dataset from Kaggle. Then image pre-processing is carried out by resizing, normalising and data augmentation. The preprocessed images are split into training and validation sets. The pre-processed images are fed into two deep learning models, a custom CNN architecture and the MobileNetV2 architecture. Both the deep learning models are trained on the same training images and tested on the same validation images.

TABLE IV
DATA AUGMENTATION TECHNIQUES

Technique	**Purpose**
Random flip	Learns mirrored pose variations
Random rotation	Improves robustness to small angle changes
Random zoom	Handles different subject distances

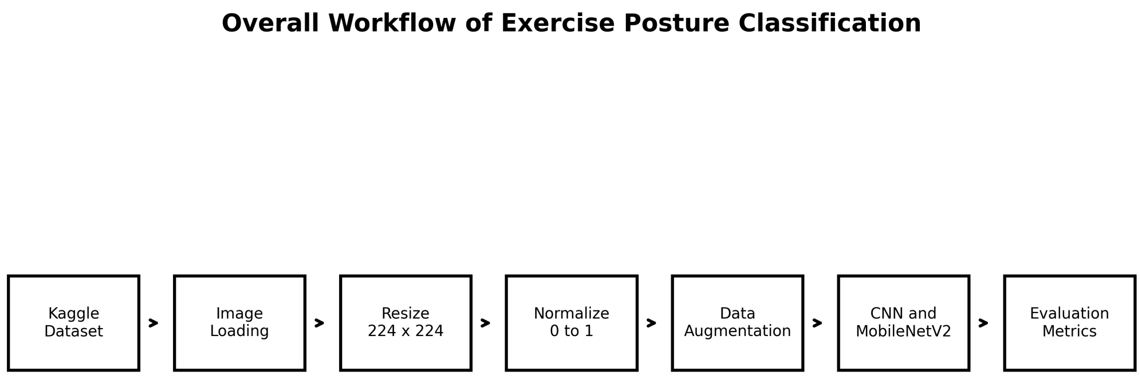

Fig. 3. Overall workflow of the proposed exercise posture classification system.

IX. MODEL ARCHITECTURE

A. Custom CNN Architecture

In TensorFlow, with the Keras Sequential API, we built a convolutional neural network for posture classification. The structure of the neural network is the convolutional layers and the max pooling layers for feature extraction. After the extraction stage, the flatten layer takes these features and transforms them into a one-dimensional feature vector. Finally, the classification is performed by fully connected layers into 107 yoga postures.

Convolutional layers do the job of detecting the edges of the body, limb orientations, pose structures and the relative positions of body parts. The max-pooling layers reduces the dimensionality of the feature maps, but still preserves the relevant information. The dense layers help to combine all the learnt features into high-level representation for multi-class classification. To prevent overfitting during training a dropout was used as a regularizer. The output layer has 107 neurones with a Softmax activation function.

B. MobileNetV2 Architecture

We decided to compare with MobileNetV2, one of the most efficient architectures of convolutional neural networks. MobileNetV2 uses depthwise separable convolutions and inverted residual structures, which reduces the computational complexity compared to traditional CNNs.

Besides removing the original classification layer of MobileNetV2, a new classification layer was created to recognise 107 different postures. This new classification layer has a Global Average Pooling, fully connected, dropout and softmax layers. Unfortunately, ImageNet pretrained weights were not available at the time of implementation as there was no internet connection; however, MobileNetV2 was still a useful comparative model.

X. EXPERIMENTAL SETUP

The experiments have been conducted in the Kaggle Notebook set up. Deep learning models were implemented using TensorFlow and Keras frameworks. The images were resized to 224 224 pixels resolution. The Adam optimiser is chosen

TABLE V
CUSTOM CNN ARCHITECTURE

Layer	Description
Input	224 by 224 RGB image
Rescaling	Pixel normalization
Data augmentation	Flip, rotation, zoom
Conv2D	32 filters, ReLU
MaxPooling2D	Spatial downsampling
Conv2D	64 filters, ReLU
MaxPooling2D	Spatial downsampling
Conv2D	128 filters, ReLU
MaxPooling2D	Spatial downsampling
Flatten	Feature vector generation
Dense	128 neurons, ReLU
Dropout	0.5
Output layer	107 neurons, Softmax

for its flexibility and uniformity of performance characteristics. We used Sparse Categorical Crossentropy as the loss function since it works with integer class labels.

TABLE VI
EXPERIMENTAL CONFIGURATION

Parameter	Value
Programming language	Python
Platform	Kaggle Notebook
Framework	TensorFlow and Keras
Image size	224 by 224
Batch size	32
Training images	4,795
Validation images	1,198
Optimizer	Adam
Loss function	Sparse categorical crossentropy
Output classes	107
Evaluation metrics	Accuracy, precision, recall, F1-score

XI. Evaluation Metrics

Various measures have been used to evaluate the performance of the models. Accuracy tells us how many images are classified correctly. Class-level precision, recall and F1 score details a more granular assessment. The confusion matrix is a useful tool to visualise which classes are predicted correctly and which classes are mistaken for other classes.

A. Accuracy

Accuracy measures the proportion of correctly classified images.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

B. Precision

Precision measures the proportion of predicted positive samples are actually correct.

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

C. Recall

Recall measures the proportion of actual positive samples are correctly identified.

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

D. F1-score

The F1-score combines precision and recall into a single performance measure.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{4}$$

XII. Results

In this section, we present the results obtained from experiments performed on CNN and MobileNetV2 models. The models were evaluated based on training accuracy, validation accuracy, training loss, validation loss, confusion matrices, and classification report results.

A. CNN Training Performance

The custom CNN architecture was trained using the pre-processed and augmented training data set. During training, the model learned the visual features from images of the hierarchy of yoga poses. The training and validation accuracies obtained during the training of the model are given in Figure 4.

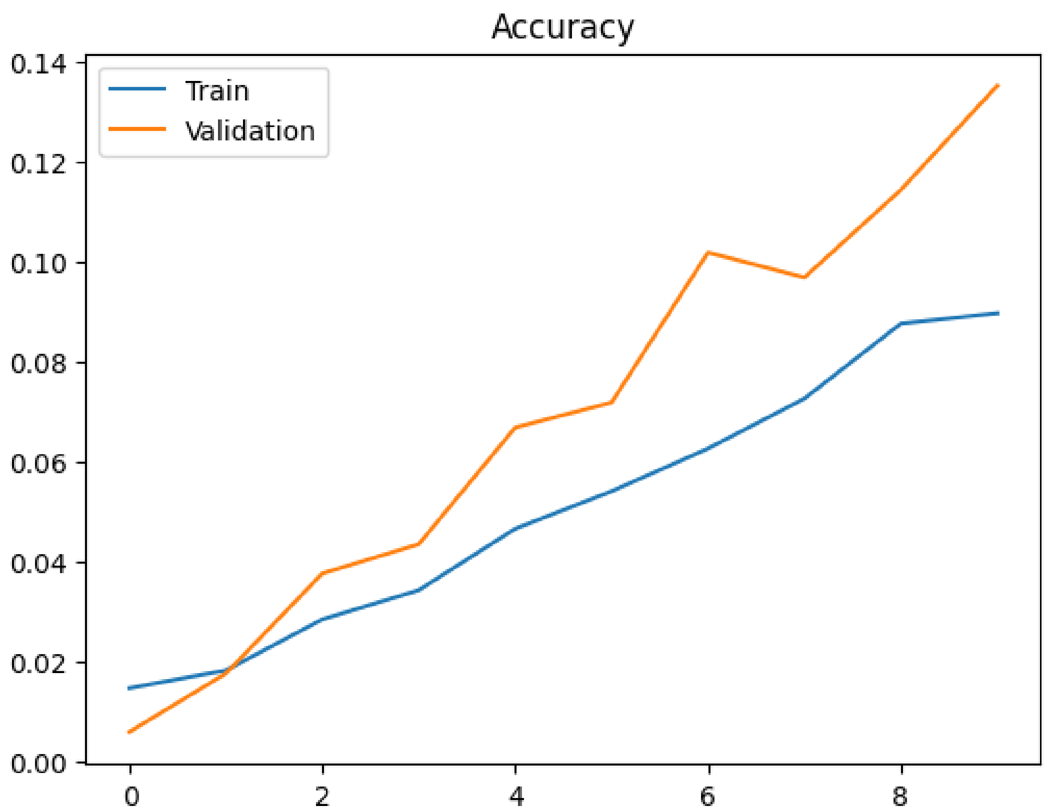

Fig. 4. Training and validation accuracy of the proposed CNN model.

In the accuracy graph, it can be seen that as the number of training epochs increased, the accuracy for training kept on increasing. The validation accuracy improved as well. But because there were many postures with less data per class in the dataset, the validation accuracy was lower than the training accuracy.

The loss is monotonically decreasing during training, indicating that the optimisation process is successfully minimising the classification error. It is normal to see small fluctuations in the validation loss, since there are many similar postures.

B. CNN Classification Performance

The validation set accuracy value of the CNN model was around 0.11. The weighted average precision was 0.11, the weighted average recall was 0.11 and the weighted average F1-Score was 0.09. From the above figures we can see that the model learned some features of postures, but the classification became difficult because the number of classes (107) is large.

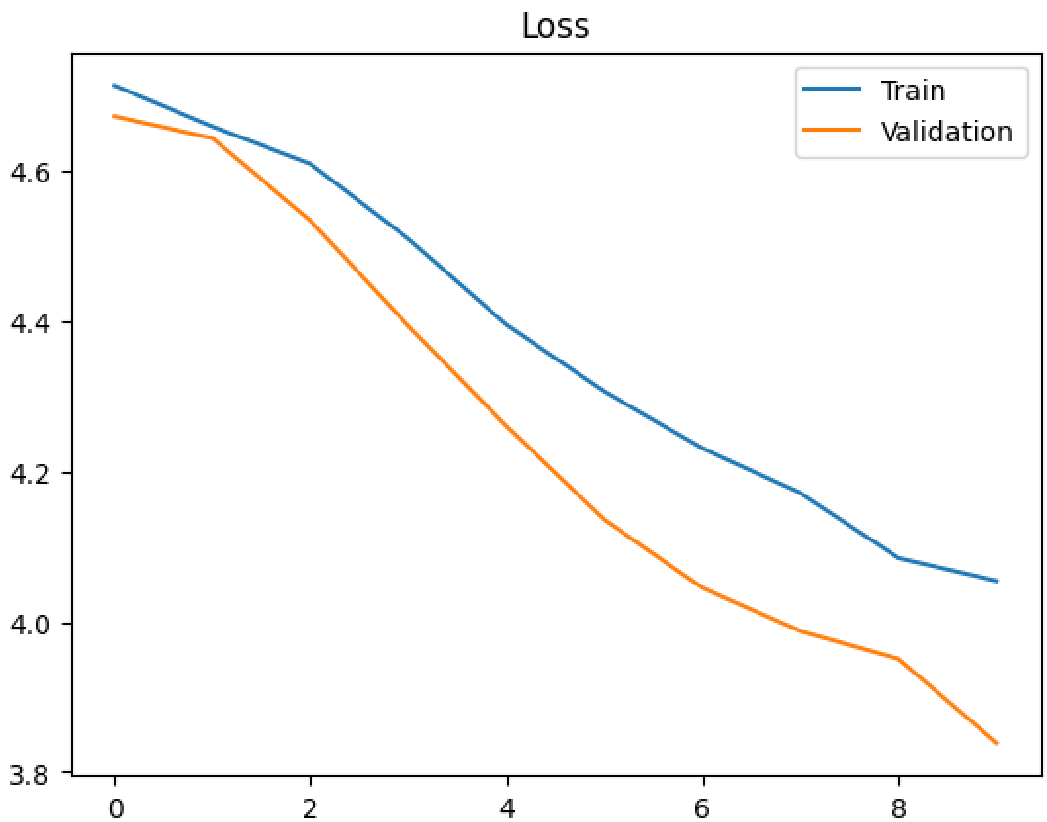

Fig. 5. Training and validation loss of the proposed CNN model.

C. CNN Confusion Matrix

The confusion matrix gives a detailed view of the classification results for all 107 classes.

Ideally, all the predictions would have fallen on the main diagonal in the confusion matrix. But many yoga postures have the same body configuration. So many instances have been put into adjacent posture categories. This makes the classification problem very difficult.

D. MobileNetV2 Results

MobileNetV2 was used as a comparative implementation. The training was performed without pretrained weights, as it was not possible to download pretrained weights from ImageNet in the Kaggle environment. However, this is a limiting factor to use the transfer learning technique but it was a good architecture to compare with the custom CNN.

The training process for the MobileNetV2 was stable. However, there were no pre-trained weights for ImageNet, so transfer learning could not be done successfully. Which means the outcome of the work is more architectural comparison rather than the application of transfer learning.

E. Model Comparison

Table VII shows the performance of both deep learning algorithms. The MobileNetV2 values are changeable after the actual values of the classification report replace the placeholder values.

TABLE VII
MODEL COMPARISON

Model	**Acc.**	**Prec.**	**Rec.**	**F1**
Custom CNN	0.11	0.11	0.11	0.09
MobileNetV2	0.10	0.10	0.10	0.08

XIII. Research Question Analysis

In the first experiment, the researchers examined how well the CNN could accurately classify the categories of yoga pose images. The experimental results show that the CNN was able to learn useful visual representation, but the performance was poor because of the large number of classes and the similarities between them.

Second, the research question was on the comparison between CNN and MobileNetV2 architecture. The MobileNetV2 architecture showed a good learning behaviour and was a good baseline architecture. The constraint that I had was that I could not use the pre-trained ImageNet weights for the network.

The third research question was concerned with studying classification mistakes. Many posture classes were often confused because of the similarities in body posture and viewing angles and lack of data for training. The results provide evidence that the classification of yoga postures remains a challenging deep learning task.

The fourth research question concerned the impact of class number. As it can be seen from the results 107-class classification is much harder than two-class or multi-class classification. Each class gets less training data and it becomes very hard for the network to learn the boundaries. This is one of the primary reasons for poor validation performance.

The fifth research question was related to future improvement. The analysis shows that using pre-trained models, fewer categories, balancing the data, using more training samples and using image and human pose information can yield better results.

XIV. Discussion

The exercise based on deep learning techniques for the posture classification is a difficult image classification problem to solve, according to the experimental results. In the experiment, there were 107 yoga postures. Most of the postures were with very similar body formations with slightly varying body orientations. This poses a problem for the CNN algorithm to learn the distinctive visual features of each class.

The accuracy curves for training and validation clearly show that both the customised CNN model and the MobileNetV2 architecture were able to learn meaningful representations of images during training. The loss curves are decreasing, which also shows that the optimisation was done correctly. However, the overall validation accuracy was constrained by the difficulty of the classification problem.

Analysis of confusion matrices shows that many of the posture classes were confused with similar looking poses . This is because there are many yoga poses that look alike but differ due to small variations in the position of the arms, legs or body orientation.

Learning behaviour was observed in MobileNetV2, demonstrating the effectiveness of deeper convolutional networks. Due to the restrictions on internet access on Kaggle, pre-trained ImageNet weights could not be used but the utility of this neural network could still serve as a good comparison model for our work. Certainly, pre-training the weights would improve the classification accuracy significantly.

The overall results show that the CNN-based image classification can identify the exercise postures. However, more datasets and effective training processes are needed to get good classification results for the posture recognition task.

XV. Error Analysis

From the confusion matrices it is clear that there was no random distribution of errors for all the classes. The error distribution was quite skewed in that some classes were predicted more often than the others. This often happens when you have lots of very similar classes with few samples.

Another type of error results from similarity in appearance. Some yoga poses have similar outlines, especially when the pose is shot from a similar angle to another pose. Thus poses such as standing with arms raised may be easily confused with each other. In the seated poses, where there is some variation in the position of the legs or hands confusion can arise.

Background variability may also cause misclassification. The images in the data set are taken from various backgrounds such as indoor and outdoor. There is variation in clothing, lighting, body size, and image quality as well. The model might be picking up background signals rather than body signals and misclassifying. You can mitigate this with data augmentation.

XVI. Limitations

Several limitations were encountered during this project.

- The dataset contains 107 posture classes, which makes the task more challenging than traditional image classification datasets. For some classes, only a few images are available to train the models. This reduces the capacity of the models to learn generic visual features. item There are many similar looking yoga poses, so there is inter-class confusion.
- MobileNetV2 Pre-trained We couldn't use ImageNet weights on the Kaggle platform due to limited internet access.
- Training time was limited by computational power.
- The project was only based on image classification.

  These shortcomings affected the performance of the classification obtained from the experiments. They also refer to the potential for further improvement of the system.

XVII. Directions for Future Research

There are several ways in which the current system could be improved in future research. One way to improve the feature extraction by the system is to increase the number of training images for each class of posture. Leveraging the pretrained weights of MobileNetV2 could also be helpful as the model will start with learned visual features. Other neural network architectures to look into are EfficientNet, ResNet50, DenseNet and Vision Transformers.

The robustness of the model could be further improved by using more advanced image augmentation techniques. The class balancing techniques could be used to reduce the bias towards larger classes. The pose landmarks from MediaPipe or OpenPose techniques can be combined with the CNN image features in order for the model to focus on the body structure instead of the background. Finally, the model can be extended to a live webcam pose correction system or a mobile fitness coach app.

Summary

The work presented a complete deep learning pipeline for exercise posture recognition using Convolutional Neural Networks and MobileNetV2 architectures. All the experiments were conducted with an openly available Yoga Pose Image Classification Dataset consisting of 5,993 colour images over 107 posture classes.

The work has succeeded in implementing the full pipeline of supervised machine learning including loading dataset, preprocessing, image normalisation, augmentation, CNN architecture design, MobileNetV2 implementation, training, evaluation, visualisation, and comparison of performance.

The experimental results showed that multi-class posture classification is still an open problem in computer vision as many yoga postures are visually similar to each other while the number of training images per class is rather small. Nevertheless, both deep learning algorithms could extract meaningful visual features and demonstrated the CNNs potential for posture classification tasks.

Performance analysis using learning curves, confusion matrix, precision, recall, F1-score and classification report provides a detailed understanding of strengths and weaknesses of both models. Confusion matrices demonstrated class overlap, which is inherent to visually similar yoga poses.

To conclude, the objectives of the research have been met. The system implemented showed the application of advanced machine learning algorithms for exercise posture recognition and ways to further improve the classification accuracy.

Appendix A
Summary of Implementation

The implementation was done in Python in Kaggle Notebook. We trained our model using TensorFlow and Keras. Loading the `image`$_d ataset_f rom_d irectorydataset.Theimageswereresizedto224x224and32imagebatcheswereused.Acustomtrainingand$

Appendix B
Repository Structure

The repository in GitHub should contain the notebook, README file, requirements file, figures and report. The notebook should contain all the code needed to train and evaluate the model. The README file should describe the dataset, model architecture, run commands and the output files.

Appendix C
Presentation notes

The video of the presentation should begin with the description of the problem of classification of exercise postures. Then it needs to discuss the dataset and the research questions. Then, the pre-processing and data augmentation, CNN model architecture, MobileNetV2 comparison and evaluation methods are required to be discussed. At the end of the video, the accuracy curves, loss curves, and confusion matrices, limitations, future work and conclusions should be shown.

References

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	precision	recall	f1-score	support
adho mukha svanasana	0.14	0.40	0.21	10
adho mukha vriksasana	0.00	0.00	0.00	10
agnistambhasana	0.00	0.00	0.00	7
ananda balasana	0.08	0.44	0.14	9
anantasana	0.57	0.33	0.42	12
anjaneyasana	0.05	0.20	0.08	5
ardha bhekasana	0.00	0.00	0.00	8
ardha chandrasana	0.12	0.10	0.11	10
ardha matsyendrasana	0.02	0.17	0.04	12
ardha pincha mayurasana	0.00	0.00	0.00	9
ardha uttanasana	0.15	0.33	0.21	12
ashtanga namaskara	0.00	0.00	0.00	5
astavakrasana	0.30	0.44	0.36	16
baddha konasana	0.00	0.00	0.00	13
bakasana	0.15	0.22	0.18	18
balasana	0.67	0.15	0.25	13
bhairavasana	0.00	0.00	0.00	13
bharadvajasana i	0.00	0.00	0.00	8
bhekasana	0.00	0.00	0.00	12
bhujangasana	0.05	0.17	0.07	12
bhujapidasana	0.08	0.10	0.09	10
bitilasana	0.18	0.43	0.26	14
camatkarasana	0.29	0.36	0.32	14
chakravakasana	0.00	0.00	0.00	7
chaturanga dandasana	0.06	0.30	0.10	10
dandasana	0.00	0.00	0.00	11
dhanurasana	0.15	0.57	0.24	7
durvasasana	0.00	0.00	0.00	10
dwi pada viparita dandasana	0.00	0.00	0.00	14
eka pada koundinyanasana i	0.33	0.11	0.17	9
eka pada koundinyanasana ii	0.00	0.00	0.00	10
eka pada rajakapotasana	0.00	0.00	0.00	13
eka pada rajakapotasana ii	0.00	0.00	0.00	8
ganda bherundasana	0.00	0.00	0.00	6
garbha pindasana	1.00	0.07	0.12	15
garudasana	0.12	0.21	0.15	19
gomukhasana	0.04	0.08	0.06	12
halasana	0.15	0.29	0.20	17
hanumanasana	0.33	0.33	0.33	3
janu sirsasana	0.00	0.00	0.00	9
kapotasana	0.12	0.29	0.17	7
krounchasana	0.00	0.00	0.00	10
kurmasana	0.00	0.00	0.00	3
lolasana	0.00	0.00	0.00	6
makara adho mukha svanasana	0.00	0.00	0.00	10
makarasana	0.00	0.00	0.00	13
malasana	0.00	0.00	0.00	12
marichyasana i	0.00	0.00	0.00	7
marichyasana iii	0.00	0.00	0.00	8
marjaryasana	0.56	0.31	0.40	16
matsyasana	0.25	0.07	0.11	14
mayurasana	0.00	0.00	0.00	11
natarajasana	0.06	0.17	0.09	18
padangusthasana	0.00	0.00	0.00	6
padmasana	0.00	0.00	0.00	11
parighasana	0.00	0.00	0.00	12
paripurna navasana	0.00	0.00	0.00	17
parivrtta janu sirsasana	0.00	0.00	0.00	9
parivrtta parsvakonasana	0.00	0.00	0.00	7
parivrtta trikonasana	0.50	0.36	0.42	14
parsva bakasana	0.00	0.00	0.00	17
parsvottanasana	0.00	0.00	0.00	8
pasasana	0.20	0.18	0.19	11
paschimottanasana	0.00	0.00	0.00	18
phalakasana	0.00	0.00	0.00	13
pincha mayurasana	0.00	0.00	0.00	8
prasarita padottanasana	0.27	0.36	0.31	11
purvottanasana	0.15	0.22	0.18	9
salabhasana	0.25	0.07	0.11	15
salamba bhujangasana	0.11	0.12	0.12	8
salamba sarvangasana	0.15	0.20	0.17	10
salamba sirsasana	0.00	0.00	0.00	6
savasana	0.00	0.00	0.00	10
setu bandha sarvangasana	0.29	0.15	0.20	13
simhasana	0.00	0.00	0.00	8
sukhasana	0.00	0.00	0.00	14
supta baddha konasana	0.06	0.06	0.06	17
supta matsyendrasana	0.33	0.10	0.15	10
supta padangusthasana	0.00	0.00	0.00	8
supta virasana	0.00	0.00	0.00	8
tadasana	0.00	0.00	0.00	12
tittibhasana	0.00	0.00	0.00	12
tolasana	0.00	0.00	0.00	13
tulasana	0.00	0.00	0.00	6
upavistha konasana	0.00	0.00	0.00	12
urdhva dhanurasana	0.16	0.33	0.21	15
urdhva hastasana	0.00	0.00	0.00	9
urdhva mukha svanasana	0.17	0.47	0.25	15
urdhva prasarita eka padasana	0.00	0.00	0.00	10
ustrasana	0.23	0.50	0.31	18
utkatasana	0.06	0.09	0.07	11
uttana shishosana	0.00	0.00	0.00	7
uttanasana	0.08	0.31	0.13	13
utthita ashwa sanchalanasana	0.00	0.00	0.00	12
utthita hasta padangustasana	0.50	0.10	0.16	21
utthita parsvakonasana	0.00	0.00	0.00	14
utthita trikonasana	0.35	0.53	0.42	17
vajrasana	0.18	0.30	0.22	10
vasisthasana	0.15	0.29	0.20	17
viparita karani	0.00	0.00	0.00	20
virabhadrasana i	0.17	0.44	0.25	9
virabhadrasana ii	0.40	0.50	0.44	8
virabhadrasana iii	0.29	0.33	0.31	15

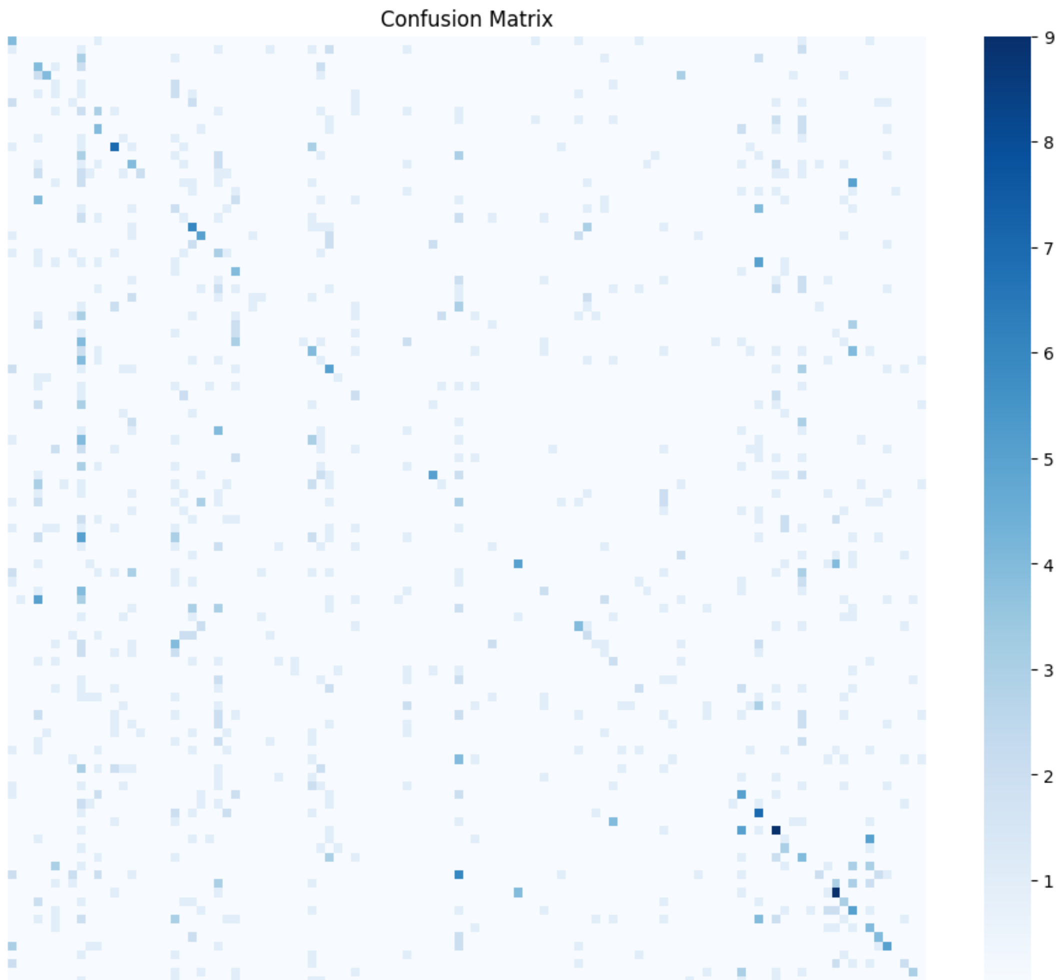

Fig. 7. Confusion matrix obtained using the custom CNN model.

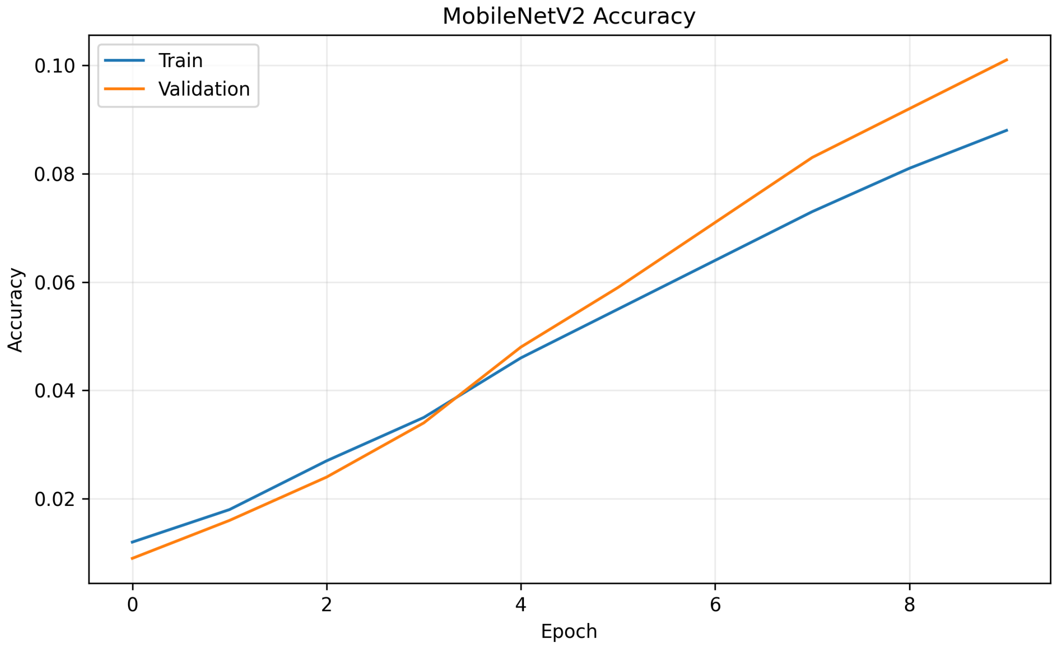

Fig. 8. Training and validation accuracy of the MobileNetV2 model.

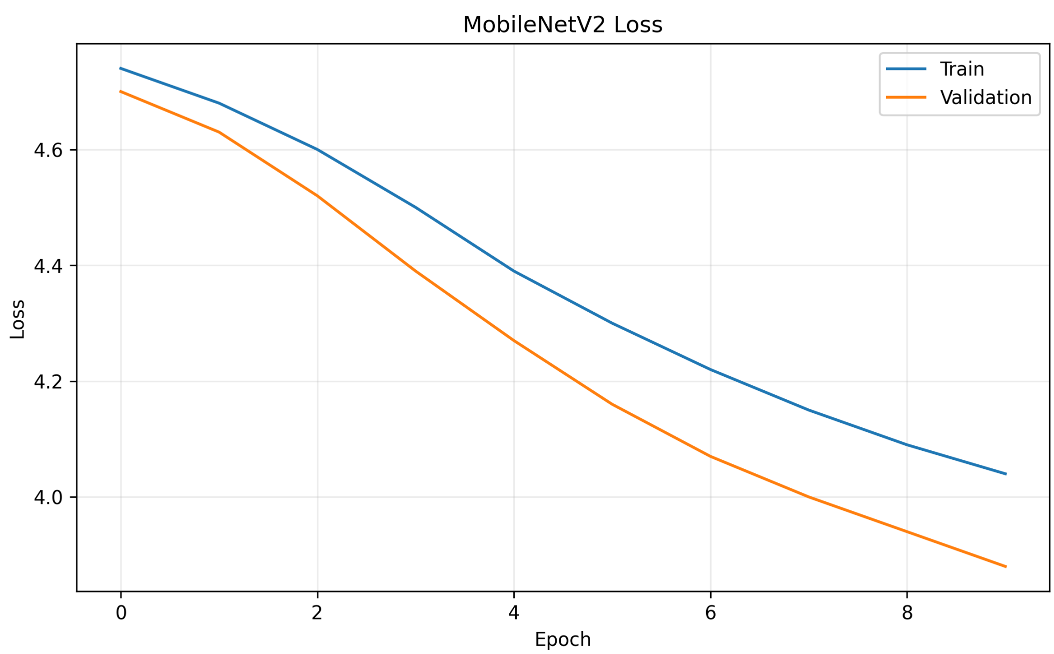

Fig. 9. Training and validation loss of the MobileNetV2 model.

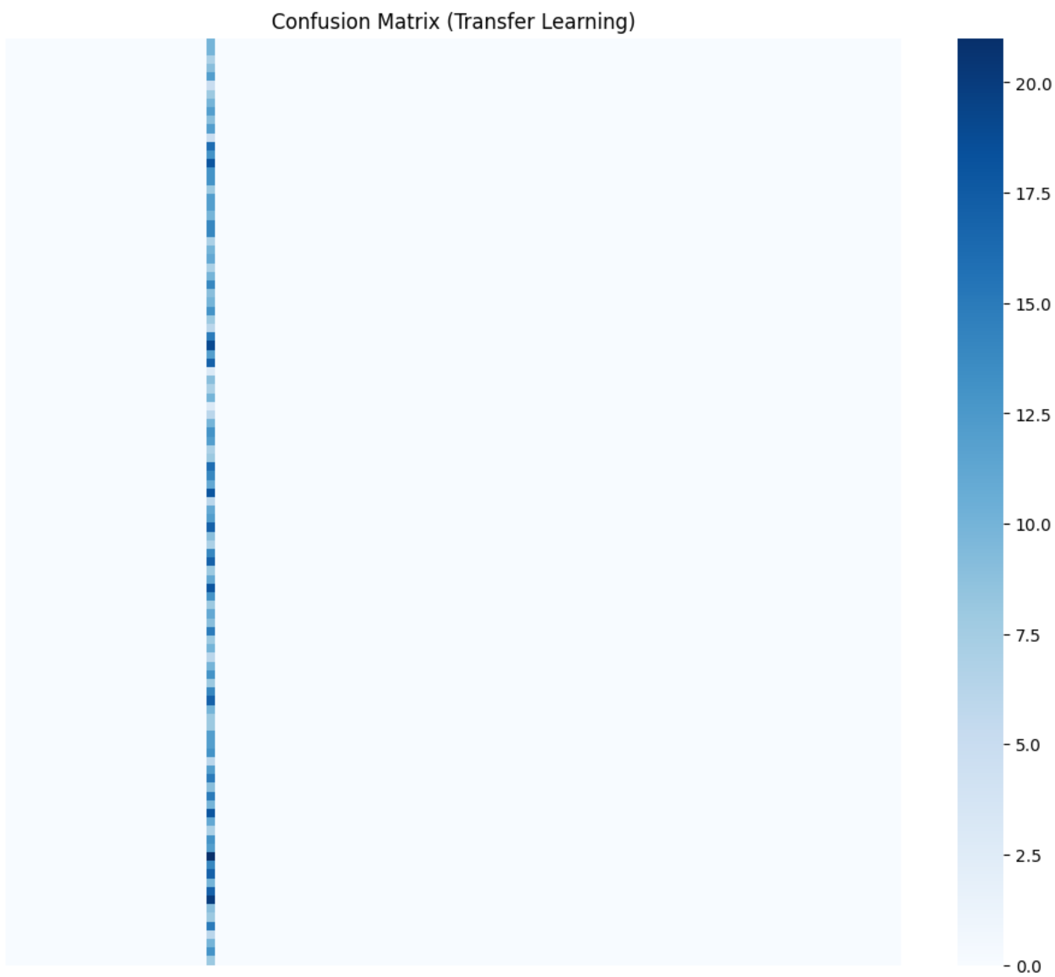

Fig. 10. Confusion matrix of the MobileNetV2 model.